

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	E Ajay Velmail	Reg. No.:	192311385
Date:		Duration:	60 minutes

Code of Conduct

I Ajay Velmail E (192311385) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ E Ajay Velmail _____

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15 %)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25 %)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied

	Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition			
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Cold-Chain Logistics Integrity Platform – Test Case Design

1. Functional Requirements

Req. ID	Functional Requirement
FR01	Register and authenticate authorized users.
FR02	Connect and manage IoT sensors/devices.
FR03	Collect real-time temperature data.
FR04	Collect humidity data.
FR05	Track GPS location of shipments.
FR06	Monitor refrigerated-truck/container door status.
FR07	Monitor refrigeration-unit health.
FR08	Configure product-specific environmental thresholds.
FR09	Detect temperature/humidity excursions automatically.
FR10	Generate real-time alerts and notifications.
FR11	Predict possible shipment/refrigeration failures.
FR12	Display real-time shipment status on a dashboard.
FR13	Store telemetry and historical shipment data.
FR14	Generate compliance reports.
FR15	Support multiple shipments and thousands of connected vehicles.
FR16	Maintain audit/history records of alerts and actions.
FR17	Recover from sensor/network/system failures without losing critical data.
FR18	Securely transmit sensor and shipment information.

2. Non-Functional Requirements

NFR ID	Requirement
NFR01	Scalability: Support thousands of simultaneously connected vehicles/sensors.
NFR02	Availability: Platform should remain operational during component failures.
NFR03	Performance: Telemetry and alerts should be processed with minimal delay.
NFR04	Reliability: Sensor readings and alerts should not be lost during temporary failures.
NFR05	Security: Data must be securely transmitted and access must be restricted.
NFR06	Usability: Dashboard and alerts should be easy for logistics operators to understand.
NFR07	Integrity: Recorded telemetry and compliance reports must remain accurate and tamper-resistant.
NFR08	Maintainability: Thresholds and device configurations should be easily updated.

3. Test Scenarios

1. User login and authentication.
2. IoT device registration.
3. Real-time temperature ingestion.
4. Real-time humidity ingestion.
5. GPS location tracking.
6. Door-open/door-closed monitoring.
7. Refrigeration-unit health monitoring.
8. Product threshold configuration.
9. Temperature excursion detection.
10. Humidity excursion detection.
11. Real-time alert generation.
12. Alert acknowledgement.
13. Shipment status monitoring.
14. Failure prediction.
15. Network interruption handling.
16. Sensor failure handling.
17. Power/refrigeration failure handling.
18. Historical telemetry retrieval.
19. Compliance report generation.
20. Unauthorized access prevention.
21. Invalid sensor data handling.
22. Large-scale concurrent device connections.
23. Data recovery after system failure.
24. Audit-log generation.
25. Dashboard usability and responsiveness.

4. Detailed Test Cases

Assumption for test data: Example pharmaceutical product has an allowed temperature range of 2°C–8°C. The exact threshold would be configurable for each product.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
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TC01	Valid user login	Enter valid username/password and login	Valid credentials	User successfully logs in and dashboard is displayed	To be executed	Not Run
TC02	Invalid login	Enter incorrect password	Username + wrong password	Login is rejected and error message displayed	To be executed	Not Run
TC03	Register IoT sensor	Add a new sensor with valid device details	Device ID: S001	Sensor is successfully registered	To be executed	Not Run
TC04	Duplicate sensor registration	Register an already registered sensor	Device ID: S001	System rejects duplicate device	To be executed	Not Run
TC05	Valid temperature reading	Send normal temperature telemetry	5°C	Reading is stored and displayed as normal	To be executed	Not Run
TC06	Minimum temperature boundary	Send lower threshold	2°C	Reading is accepted and marked within allowed range	To be executed	Not Run
TC07	Maximum temperature boundary	Send upper threshold	8°C	Reading is accepted and marked within allowed range	To be executed	Not Run
TC08	Temperature below limit	Send temperature below allowed range	1.9°C	Excursion is detected and alert generated	To be executed	Not Run
TC09	Temperature above limit	Send temperature above allowed range	8.1°C	Excursion is detected and alert generated	To be executed	Not Run
TC10	Invalid temperature	Send invalid/non-numeric value	"ABC"	Invalid reading is rejected/logged and system	To be executed	Not Run

				continues operating		
TC1 1	Humidity monitoring	Send valid humidity reading	55% RH	Humidity is displayed and stored correctly	To be executed	Not Run
TC1 2	Humidity boundary	Send configured humidity limit	60% RH	Reading is handled according to configured threshold	To be executed	Not Run
TC1 3	GPS tracking	Send valid GPS coordinates	Latitude/longitude	Shipment location is updated correctly	To be executed	Not Run
TC1 4	Invalid GPS data	Send invalid coordinates	Latitude 200°	Invalid location is rejected and logged	To be executed	Not Run
TC1 5	Door closed status	Send door status	CLOSED	Dashboard shows door as closed	To be executed	Not Run
TC1 6	Door opened	Change door status to OPEN	OPEN	Dashboard updates and event is recorded	To be executed	Not Run
TC1 7	Prolonged door opening	Keep door open beyond configured duration	OPEN for 10 min	Alert is generated if configured limit is exceeded	To be executed	Not Run
TC1 8	Refrigeration failure	Simulate refrigeration-unit failure	Compressor = FAILED	Failure is detected and responsible users are alerted	To be executed	Not Run
TC1 9	Alert generation	Create temperature excursion	10°C	Immediate alert is sent to configured stakeholders	To be executed	Not Run
TC2 0	Alert acknowledgment	Open and acknowledge an alert	Alert ID A001	Alert becomes acknowledged and action is recorded	To be executed	Not Run

TC2 1	Threshold configuration	Change product temperature range	2–8°C → 2–10°C	New threshold is saved and applied to future readings	To be executed	Not Run
TC2 2	Invalid threshold	Configure minimum greater than maximum	Min = 10°C, Max = 2°C	System rejects configuration	To be executed	Not Run
TC2 3	Predictive failure	Simulate degrading refrigeration performance	Increasing temperature + unit warnings	System identifies increased failure risk and generates prediction/alert	To be executed	Not Run
TC2 4	Network interruption	Disconnect vehicle from network temporarily	Network unavailable for 5 min	System handles outage without crashing; buffered data is recovered if supported	To be executed	Not Run
TC2 5	Sensor failure	Stop temperature sensor transmission	No telemetry	Sensor failure/stale data is identified and appropriate alert generated	To be executed	Not Run
TC2 6	Historical data	Request previous shipment telemetry	Shipment S001, previous 24 hours	Correct historical readings are displayed	To be executed	Not Run
TC2 7	Compliance report	Generate report after shipment completion	Shipment S001	Complete report containing environmental history and excursions is generated	To be executed	Not Run

TC28	Unauthorized report access	Attempt to access report without permission	Unauthorized user	Access is denied	To be executed	Not Run
TC29	Concurrent devices	Connect a large number of simulated vehicles	Thousands of devices	System remains available and processes telemetry without unacceptable degradation	To be executed	Not Run
TC30	Data recovery	Restart/fail a platform component during telemetry ingestion	Active shipment telemetry	System recovers and critical telemetry is not incorrectly lost	To be executed	Not Run

5. Equivalence Partitioning

Equivalence Partitioning divides input data into groups that should behave similarly.

Temperature

For a product with a valid range of 2°C–8°C:

Partition	Input	Expected
EP1 – Below range	< 2°C	Excursion alert
EP2 – Valid range	2°C–8°C	Normal
EP3 – Above range	> 8°C	Excursion alert
EP4 – Invalid	Text/non-numeric	Reject/log invalid data

Example test cases: **TC05, TC06, TC07, TC08, TC09, TC10.**

6. Boundary Value Analysis

Boundary Value Analysis tests values immediately around limits.

For temperature 2°C–8°C:

Test	Temperature	Expected
BVA1	1.9°C	Alert
BVA2	2.0°C	Normal
BVA3	2.1°C	Normal
BVA4	7.9°C	Normal
BVA5	8.0°C	Normal
BVA6	8.1°C	Alert

This verifies that the system correctly handles the exact configured limits.

7. Decision Table Testing

Consider a shipment where an alert should be generated when temperature exceeds its threshold or the refrigeration unit fails.

Condition	Rule 1	Rule 2	Rule 3	Rule 4
Temperature within range	Yes	No	Yes	No
Refrigeration working	Yes	Yes	No	No
Generate critical alert	No	Yes	Yes	Yes
Mark shipment at risk	No	Yes	Yes	Yes

Example

- **Rule 1:** Temperature normal + refrigeration working → shipment normal.
- **Rule 2:** Temperature excursion + refrigeration working → temperature alert.
- **Rule 3:** Temperature normal + refrigeration failure → refrigeration alert.
- **Rule 4:** Temperature excursion + refrigeration failure → critical shipment-risk alert.

8. State Transition Testing

A shipment can move through several states:

Created → In Transit → At Risk → Alert Generated → Alert Acknowledged → Recovered/Delivered

Example:

Current State	Event	Expected Next State
Created	Shipment starts	In Transit
In Transit	Temperature exceeds limit	At Risk
At Risk	Excursion confirmed	Alert Generated
Alert Generated	Operator acknowledges	Alert Acknowledged
Alert Acknowledged	Temperature returns to normal	Recovered
In Transit	Shipment reaches destination	Delivered

State Test Cases

- Valid transition: **In Transit** → **At Risk**
- Valid transition: **At Risk** → **Alert Generated**
- Valid transition: **Alert Generated** → **Alert Acknowledged**
- Invalid transition: **Delivered** → **In Transit**
- Invalid transition: **Created** → **Delivered** without shipment processing

9. Non-Functional Test Cases

TC ID	Non-Functional Test	Test Method	Expected Result
NFT01	Performance	Send continuous telemetry from many devices	Telemetry is processed within the defined response target
NFT02	Scalability	Simulate thousands of simultaneous vehicles	Platform continues operating without system failure
NFT03	Availability	Stop one application/service component	Platform continues operating through redundancy/failover
NFT04	Reliability	Interrupt network connection	Data is recovered/processed according to system design
NFT05	Security	Attempt unauthorized dashboard access	Access is denied
NFT06	Data integrity	Compare transmitted and stored telemetry	Stored data matches valid sensor data
NFT07	Usability	Operator performs common dashboard tasks	Tasks can be completed clearly and efficiently
NFT08	Recovery	Restart a failed service	Service recovers and resumes processing
NFT09	Auditability	Modify threshold configuration	User, timestamp and configuration change are recorded

NFT10	Concurrent processing	Generate simultaneous alerts from multiple vehicles	Alerts are processed without incorrect mixing of shipment data
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10. Traceability Matrix

The assignment's rubric gives significant importance to traceability between requirements, scenarios and test cases.

Requirement	Related Test Cases	Coverage
FR01 – Authentication	TC01, TC02	✓
FR02 – IoT registration	TC03, TC04	✓
FR03 – Temperature	TC05–TC10	✓
FR04 – Humidity	TC11, TC12	✓
FR05 – GPS	TC13, TC14	✓
FR06 – Door monitoring	TC15–TC17	✓
FR07 – Refrigeration health	TC18	✓
FR08 – Threshold configuration	TC21, TC22	✓
FR09 – Excursion detection	TC08, TC09, TC19	✓
FR10 – Notifications	TC19, TC20	✓
FR11 – Failure prediction	TC23	✓
FR12 – Dashboard	TC05, TC13, TC15	✓
FR13 – Data storage	TC05, TC11, TC26	✓
FR14 – Compliance reports	TC27, TC28	✓
FR15 – Scalability	TC29	✓
FR16 – Audit trail	TC20, TC21	✓
FR17 – Fault tolerance	TC24, TC25, TC30	✓
FR18 – Secure transmission/access	TC28, NFT05	✓
NFR01 – Scalability	NFT02, NFT10	✓
NFR02 – Availability	NFT03	✓
NFR03 – Performance	NFT01	✓
NFR04 – Reliability	NFT04, NFT08	✓
NFR05 – Security	NFT05	✓
NFR06 – Usability	NFT07	✓
NFR07 – Integrity	NFT06	✓
NFR08 – Maintainability	TC21, TC22	✓

11. Test Coverage Summary

Test Category	Number of Tests
Functional test cases	30
Non-functional test cases	10
Boundary-value tests	6
Equivalence partitions	4
Decision-table rules	4
State-transition cases	5
Total planned test activities	59

Coverage

- **Functional requirements:** Covered
- **Non-functional requirements:** Covered
- **Normal conditions:** Covered
- **Boundary conditions:** Covered
- **Invalid conditions:** Covered
- **Exceptional/failure conditions:** Covered
- **Security:** Covered
- **Scalability:** Covered
- **Fault tolerance:** Covered
- **Compliance reporting:** Covered
- **Traceability:** Covered

The assignment's evaluation criteria specifically reward comprehensive coverage, precise test data/expected results, application of multiple test-design techniques, traceability, and professional documentation.

12. Conclusion

The proposed test suite verifies the Cold-Chain Logistics Integrity Platform from both **functional and quality perspectives**. It tests real-time IoT telemetry, environmental threshold monitoring, GPS tracking, door and refrigeration status, alerts, predictive failure detection, compliance reporting, security, scalability, reliability and recovery.

Using **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing** provides stronger coverage than testing only normal inputs. This

approach is aligned with the assessment requirement to apply appropriate test-design techniques and demonstrate adequate requirement and functional coverage.